Example 2.21 A Kho-Kho ground has dimensions 30 $m \times 19$ m which includes a lobby on all of its sides. The dimensions of the playing area is 27 $m \times 16$ m. Find the area of the lobby.

Solution

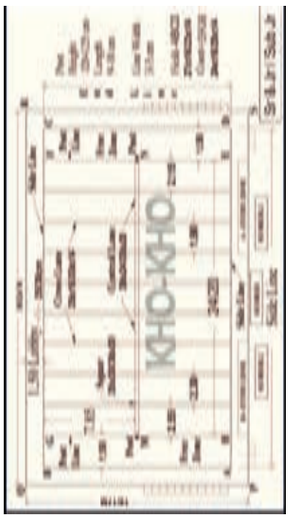


Fig. 2.25

From the dimensions of the ground we have,

$L = 30\ m$; $B = 19\ m$; $l = 27\ m$; $b = 16\ m$

Area of the kho kho ground $= L \times B$

$= 30 \times 19$

$= 570\ m^2$

Area of the play field $= l \times b$

$= 27 \times 16$

$= 432\ m^2$

Area of the lobby = Area of Kho-Kho ground − Area of the play field

$= 570 - 432$

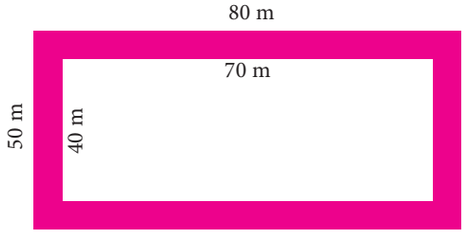
$= 148\ m^2$

Exercise 2.3

1. Find the area of a circular pathway whose outer radius is 32 *cm* and inner radius is 18 *cm*.
2. There is a circular lawn of radius 28 *m*. A path of 7 *m* width is laid around the lawn. What will be the area of the path?
3. A circular carpet whose radius is 106 *cm* is laid on a circular hall of radius 120 *cm*. Find the area of the hall uncovered by the carpet.
4. A school ground is in the shape of a circle with radius 103 *m*. Four tracks each of 3 *m* wide has to be constructed inside the ground for the purpose of track events. Find the cost of constructing the track at the rate of ₹50 per *sq.m.*



5. The figure shown is the aerial view of the pathway. Find the area of the pathway.



6. A rectangular garden has dimensions 11 $m \times 8$ m. A path of 2 *m* wide has to be constructed along its sides. Find the area of the path.

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